

Week 3

[Row Echelon Form](#)

[Reduced Row Echelon Form](#)

[Rank of a Matrix](#)

[How to find rank of a matrix?](#)

[Span of a set of vectors](#)

[Basis and Dimension](#)

[Basis \(of a column/row space\)](#)

[Conditions for \$B\$ to be a basis](#)

[Bases for \$\mathbb{R}^n\$](#)

[How to find basis of a subspace of \$\mathbb{R}^n\$?](#)

[Dimension of a Vector Space](#)

[Linear Independence](#)

[Four Fundamental Subspaces](#)

[Column Space \$C\(A\)\$](#)

[Null Space \$N\(A\)\$](#)

[How to find Null Space of \$A\$?](#)

[Row Space \$C\(A^T\)\$](#)

[Left Null Space of \$A\$](#)

[Rank-Nullity Theorem](#)

[Length of a Vector \$u\(x, y\)\$](#)

[Angle between two vectors](#)

[Orthogonal Vectors](#)

[Orthogonality and Linear Independence](#)

[Orthogonal Set of Vectors](#)

[Orthonormal Set of Vectors](#)

[Projection](#)

[Projection matrix of \$A\$](#)

[Projection of \$B\$ onto \$A\$](#)

[Calculating Error after Projection](#)

[Linear Regression](#)

[Finding the best fit line](#)

If the number of columns of a matrix is less than its rank, then inverse is not possible.

Row Echelon Form

A matrix is in row echelon form if:

- The first non-zero element in each row, called the leading entry, is 1.
- Each leading entry in a column to the right of the leading entry in the previous row.
- Rows with all zero elements, if any, are below rows having a non-zero element.

Reduced Row Echelon Form

A matrix is said to be in reduced row echelon form if:

- satisfies all conditions of row echelon form
- For a non-zero row, the leading entry in the row is the only non-zero entry in its column.

Rank of a Matrix

Let A be an $m \times n$ matrix:

- The column space of A is the subspace of $\mathbb{R}^m$ spanned by the column vectors of A .
- The row space of A is the subspace of $\mathbb{R}^n$ spanned by the row vectors of A .
- The dimension of the column space of A is the column rank of A .
- The dimension of row space of A is the row rank of A .
- Column Rank = Row Rank = Rank of A

How to find rank of a matrix?

Step 1: Reduce the matrix into row echelon form.

Step 2: The number of non-zero rows in the reduced matrix is the rank of matrix A .



The rank of a matrix is always less than or equal to the number of rows (or columns).

Span of a set of vectors

The span of a set S (of vectors) is defined as the set of all finite linear combinations of elements (vectors) of S , and denoted by $Span(S)$ i.e.

$$Span(S) = \left\{ \sum_{i=1}^n a_i v_i \in V \mid a_1, a_2, \dots, a_n \in \mathbb{R} \right\}$$

Basis and Dimension



$\text{rank}(A) = \text{number of pivot columns} = \text{dimension of } C(A) = \text{dimension of } C(A^T)$

Basis (of a column/row space)

A basis B of a vector space V is a spanning set for V is a linearly independent subset of V that spans V .



A set $S = \{u_1, u_2, \dots, u_n\}$ of vectors is a basis of V if every $u \in V$ can be written uniquely as a linear combination of the basis vectors.

- n vectors in $\mathbb{R}^n$ give basis if the $n \times n$ matrix with those columns is invertible.

Conditions for B to be a basis

1. B is linearly independent and $Span(B) = V$.
2. B is a maximal linearly independent set.
3. B is a minimal spanning set.

Bases for $\mathbb{R}^n$

- If $k < n$, then the vectors $v_1, v_2, \dots, v_k$ do not span $\mathbb{R}^n$.
- If $k > n$, then the vectors $v_1, v_2, \dots, v_k$ are linearly dependent.

- If $k = n$, then the following conditions are equivalent:
 - $\{v_1, v_2, \dots, v_k\}$ is a basis for $\mathbb{R}^n$.
 - $\{v_1, v_2, \dots, v_k\}$ is a spanning set for $\mathbb{R}^n$.
 - $\{v_1, v_2, \dots, v_k\}$ is a linearly independent set.

How to find basis of a subspace of $\mathbb{R}^n$?

- Non-zero rows in row-reduced matrix.

Suppose a subspace V of $\mathbb{R}^n$ is defined by some constraints, as follows:

$$V = \{(x_1, x_2, \dots, x_n) | f_1(x_1, x_2, \dots, x_n) = 0, f_2(x_1, x_2, \dots, x_n) = 0, \dots, f_m(x_1, x_2, \dots, x_n) = 0\} \subseteq \mathbb{R}^n$$

where f_i 's are linear equations in general.

Step 1: By solving f_i 's, we have to find out the dependent and independent variables. Suppose there are k independent and $k - n$ dependent variables.

Step 2: Choose the first independent variable, let it be x_1 and assign the values 1 to that and assign 0 to all other independent variables.

Step 3: Find out the values of all the dependent variables with respect to the assignment in step and call that vector to be v_1 .

Step 4: Carry out this process for each independent variable, to get the vectors $v_2, \dots, v_k$.

The set of vectors $\{v_1, v_2, \dots, v_k\}$ is a linearly independent set and this is a maximal linearly independent set in V , hence, forms a basis of V . Hence, dimension of V is k in this case.

Dimension of a Vector Space



- Cardinality of its bases, or
- Cardinality of maximal linearly independent set, or
- Cardinality of minimal spanning set

- Dimension of the space $\mathbb{R}^n$ is n .
- Number of non-zero rows in row-reduced matrix

Linear Independence



Vectors $x_1, x_2, \dots, x_n$ are independent if no combination gives the zero vector (except the zero vector i.e. when all $c_i = 0$).



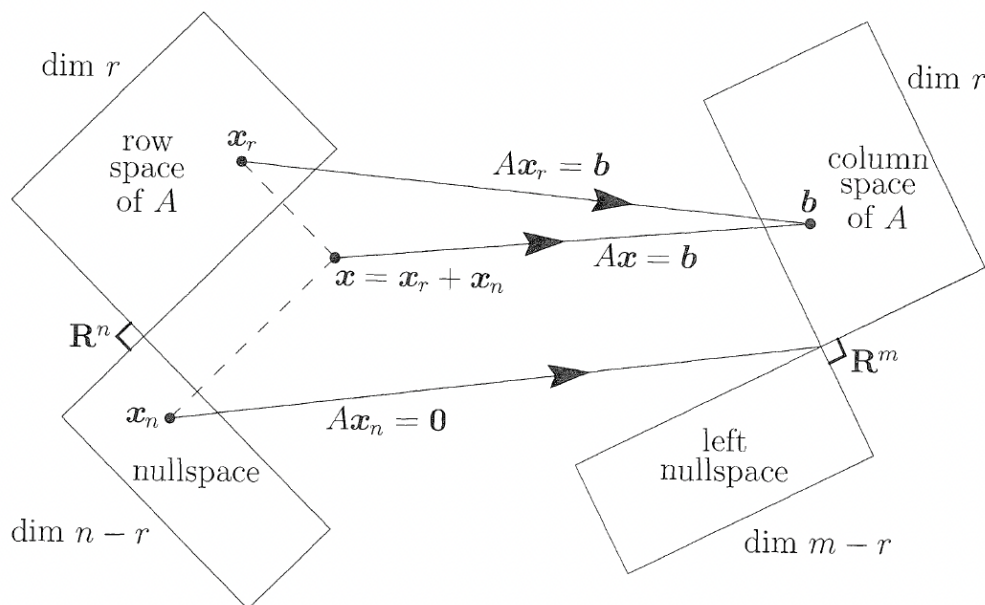
$$c_1x_1 + c_2x_2 + \dots + c_nx_n \neq 0$$

- A set of vectors $v_1, v_2, \dots, v_n$ containing the zero vector is always linear dependent.
- Two non-zero vectors are linearly independent precisely when they are not multiples of each other.
- If three vectors are linearly independent, then none of these vectors is a linear combination of the other.
- Any set of r vectors in $\mathbb{R}^n$ with $r > n$ are linearly dependent.
- If $\det(A) \neq 0$, then the set of vectors is linearly independent.

- $v_1, v_2, \dots, v_n$ are linearly independent if null space of A is the zero vector or $\text{Rank} = n$.
- $v_1, v_2, \dots, v_n$ are linearly dependent if $Ac = 0$ for some non-zero c or $\text{rank} < n$.
- Any set containing a linearly dependent set is linearly dependent.
- If a set is linearly independent, then every subset of it is linearly independent.
- If set contains the zero vector, then it is linearly dependent.
- A Singleton vector is a linear Independent set **iff** it is not the zero vector.

Four Fundamental Subspaces

Let A be $m \times n$.



Column Space $C(A)$



The set of all possible linear combinations of the columns of a matrix A .

Example:

Column Space of A / All Combinations of columns

$$Ax = \begin{bmatrix} 1 & 4 & 5 \\ 3 & 2 & 5 \\ 2 & 1 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 3 \\ 2 \end{bmatrix} x_1 + \begin{bmatrix} 4 \\ 2 \\ 1 \end{bmatrix} x_2 + \begin{bmatrix} 5 \\ 5 \\ 3 \end{bmatrix} x_3$$

= linear combination of all columns of A
= all vectors Ax

- Column Space is a subspace of $\mathbb{R}^m$

Null Space $N(A)$

- Null Space is a subspace of $\mathbb{R}^n$.

- Dimension of Null Space = $n - r$
- The dimension of the null space is called the nullity of A .

How to find Null Space of A ?

Step 1: Row reduce the matrix

Step 2: Find independent columns (which do not have pivot elements) and dependent (pivot elements are there) columns.

Step 3: Number of independent columns = Nullity of the matrix
 Number of dependent columns = Dimension of the matrix

Row Space $C(A^T)$



The set of all possible linear combinations of the rows of a matrix A .
 OR

The set of all possible linear combinations of the rows of a matrix A^T .

- Row Space is a subspace of $\mathbb{R}^n$.
- $Ax = 0$

Left Null Space of A



Null Space of $A^T = N(A^T)$

- Left Null Space is a subspace of $\mathbb{R}^m$
- $A^T y = 0$

Rank-Nullity Theorem

Let A be an $m \times n$ matrix, the

$$\text{Rank}(A) + \text{Nullity}(A) = n \text{ or } \dim(V)$$

Length of a Vector $u(x, y)$

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Angle between two vectors

$$\cos \theta = \left| \frac{u \cdot v}{\|u\| \|v\|} \right|$$

Orthogonal Vectors

- If angle θ between two vectors u and v in $\mathbb{R}^n$ is a right angle, or
- if $\langle u, v \rangle = 0$

Orthogonality and Linear Independence

If $\{v_1, v_2, \dots, v_k\}$ is an orthogonal set of vectors in the inner product space V , then $\{v_1, v_2, \dots, v_k\}$ is a linearly independent set of vectors.

Orthogonal Set of Vectors

An orthogonal set of vectors of an inner product space V is a set of vectors whose elements are mutually orthogonal.

Orthonormal Set of Vectors

An orthonormal set of vectors of an inner product space V is an orthogonal set of vectors such that the norm of each vector is 1.

Projection

Projection matrix of A

$$\mathbb{P} = \frac{AA^T}{A^T A}$$

Projection of B onto A

$$p = \mathbb{P}B$$

Calculating Error after Projection

$$e = B - p$$

Linear Regression

Finding the best fit line

We need to find

$$\hat{x} = \begin{bmatrix} \hat{\theta}' \\ \hat{\theta}'' \end{bmatrix} = \begin{bmatrix} \text{slope} \\ \text{intercept} \end{bmatrix}$$

Least Squares Method

$$A^T A \hat{x} = A^T b$$

Find $A^T A$ and $A^T b$

$$A = \begin{bmatrix} 1 & 1 \\ 2 & 1 \\ 3 & 1 \\ 4 & 1 \\ 5 & 1 \end{bmatrix} \quad b = \begin{bmatrix} 2.6 \\ 3.4 \\ 7.1 \\ 10.2 \\ 13.5 \end{bmatrix}$$
$$A^T = \begin{bmatrix} 1 & 2 & 3 & 4 & 5 \\ 1 & 1 & 1 & 1 & 1 \end{bmatrix}$$
$$A^T A = \begin{bmatrix} 1 & 2 & 3 & 4 & 5 \\ 1 & 1 & 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 2 & 1 \\ 3 & 1 \\ 4 & 1 \\ 5 & 1 \end{bmatrix} \quad A^T b = \begin{bmatrix} 1 & 2 & 3 & 4 & 5 \\ 1 & 1 & 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} 2.6 \\ 3.4 \\ 7.1 \\ 10.2 \\ 13.5 \end{bmatrix}$$
$$A^T A = \begin{bmatrix} 55 & 15 \\ 15 & 5 \end{bmatrix} \quad A^T b = \begin{bmatrix} 139 \\ 36.8 \end{bmatrix}$$
$$\begin{bmatrix} 55 & 15 \\ 15 & 5 \end{bmatrix} \begin{bmatrix} \hat{\theta}' \\ \hat{\theta}'' \end{bmatrix} = \begin{bmatrix} 139 \\ 36.8 \end{bmatrix}$$
$$55 \hat{\theta}' + 15 \hat{\theta}'' = 139$$
$$15 \hat{\theta}' + 5 \hat{\theta}'' = 36.8$$

Solving these equations;

$$\hat{\theta}' = 2.86$$
$$\hat{\theta}'' = -1.22$$

